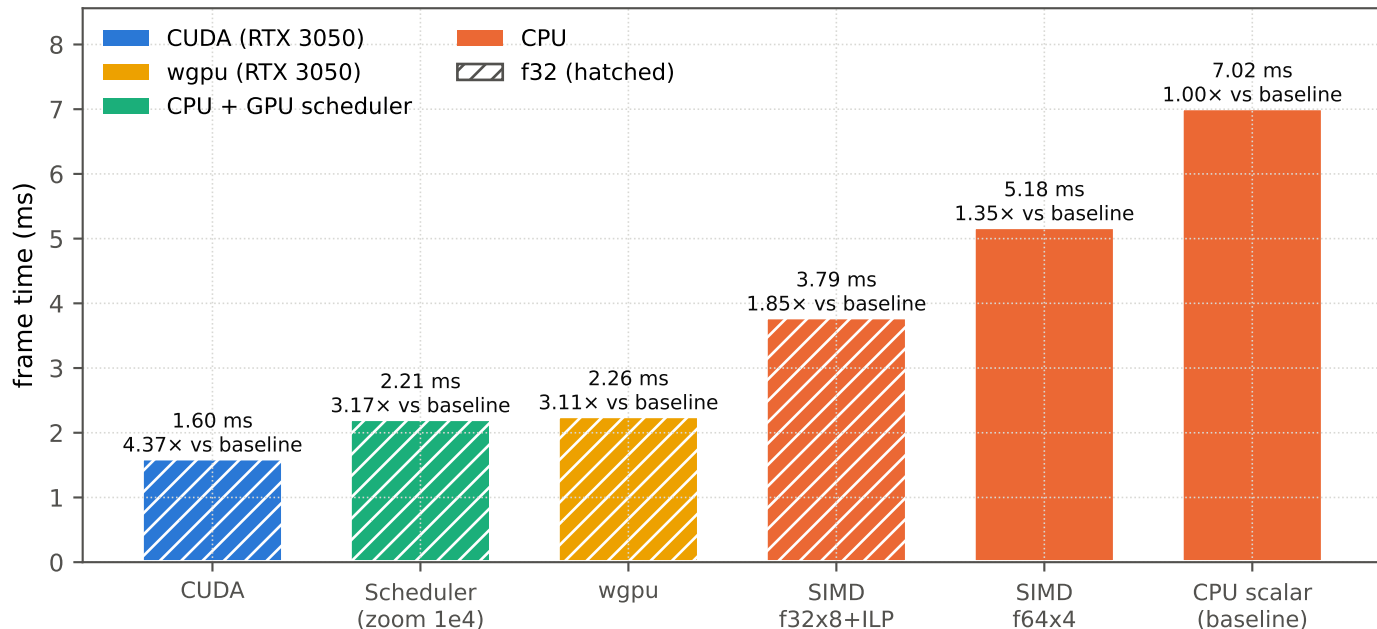


Backend comparison — 1920×1080 Mandelbrot, max_iter=1000



Hatched bars are f32, solid are f64 — not a like-for-like comparison across the two. The precision-controlled pairing is CUDA (1.61 ms) against the CPU's own best f32 path (3.79 ms) = 2.36×; against the f64 baseline it reads 4.37×.